

An introduction to Partition Theory

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Abstract

The goal of this paper is to provide an overview of partition theory, which is considered a branch of additive number theory. It begins with the definition of a partition and then explores related arithmetic functions, proving several results concerning the relationships between them. In addition, it introduces a graphical method for representing partitions, known as the Ferrers diagram, and then concluded with some remarkable result form partition theory.

1 Introduction

Additive number theory is a classical and foundational branch of number theory that focuses on expressing integers as sums of other integers. Unlike multiplicative number theory, which investigates properties like prime factorization, additive number theory is concerned with the additive structure of the integers and asks how a given number can be written as a sum of elements from a particular set—such as squares, primes, or distinct integers. The methods and answers to these problems vary depending on the nature of the summands and often require sophisticated combinatorial and analytical techniques. One of the most celebrated results in this area is Lagrange’s Four Square Theorem, proven in 1770, which asserts that every non-negative integer can be written as the sum of four integer square.

In case we allow the summands to be any positive integers, we enter the domain of **partition theory**, where the central question concerns the number of ways a given integer can be written as a sum of positive integers, regardless of order. Partition theory further allows for various restrictions on the summands—for example, requiring them to be odd, even, distinct, or belonging to specific arithmetic progressions. These conditions give rise to different types of partitions and lead to interesting identities and comparisons between them. Moreover, an important aspect of partition theory is the search for generating functions or closed-form expressions that enumerate the number of such partitions for a given integer, often revealing deep combinatorial and number-theoretic insights.

The aim of this paper is to provide an accessible introduction to partition theory. It begins with some central definitions and illustrative examples, then proceeds to discuss Ferrers diagrams and how they can be used to prove key theorems. The paper also explores one of the foundational results in partition theory—Euler’s Theorem—and concludes with an overview of several remarkable results that highlight the richness and depth of the subject.

2 Definitions

Definition 2.1. Let n be a nonnegative integer. A representation of n as a sum of positive integers is called a **partition of n** .

Example 2.1. For $n = 3$ we have $3, 2 + 1, 1 + 1 + 1$ are all possible partitions of 3 . For $n = 5$ the possible partition are $5, 4 + 1, 3 + 2, 3 + 1 + 1, 2 + 2 + 1, 2 + 1 + 1 + 1, 1 + 1 + 1 + 1 + 1$

Remarks

- *i)* The number n is itself considered a partition of n .
- *ii)* Each term in a partition of n is called a **part** of the partition of n (e.g. $1+1+1$ is a 3-parts partition of 3).
- *iii)* The order of the parts of a partition is irrelevant (e.g., $2 + 1$ and $1 + 2$ are counted as one partition of 3)
- *iv)* A partition is usually written in decreasing(not strictly) order of parts.
- *v)* The number 0 has only one partition which is the *empty partition* which has no parts.

Definition 2.2. The **partition function** $p(n)$ is a function defined as

$$p : \mathbb{N} \cup \{0\} \longrightarrow \mathbb{N}$$

where $p(n)$ = the number of partitions of n

Example 2.2. $p(0) = 1, p(3) = 3, p(5) = 7$, and $p(100) = 190,569,292$

Definition 2.3. We can obtain interesting functions by applying certain restrictions on the parts of the partitions,for n we have the following functions:

- 1) $p_m(n)$ = # partitions of n where each part is *less than or equal* m .
- 2) $p^o(n)$ = # partitions of n where each part is *is an odd number* .
- 3) $p^d(n)$ = # partitions of n where the parts are *distinct*
- 4) $q^e(n)$ = # partitions of n where the parts are *distinct and even*
- 5) $q^o(n)$ = # partitions of n where the parts are *distinct and odd*

Remarks

- *i)* By convention, $p_m(0) = p_o(0) = p^d(0) = q_e(0) = 1, q^o(0) = 0$
- *ii)* $p(n) \geq f(n)$, where $f(n)$ is any of the functions defined above.
- *iii)* if $m \geq n$, then $p(n) = p_m(n)$
- *iv)* $p^d(n) = q^o(n) + q^e(n)$

Example 2.3. For $5 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$, so $p_2(5) =$.

Also, $5 = 3 + 1 = 1 + 1 + 1 + 1 + 1$, so $p^o(5) = 3$ and $q^o(5) = 1$

Example 2.4. Show that $p_n(n) = p(n)_{n-1} + 1$ for $n > 1$.

Let $n = a_1 + a_2 + \cdots + a_r$ be a partition of n with $a_i \leq n \ \forall i \in \{1, \dots, r\}$. Now, we consider the cases :

Case 1) n is part of the partition: In this case we will have only one partition which is $n = n$.

Case 2) n is *not* part of the partition:

In this case, all the parts of the partition are less than or equal $n - 1$, thus we will have $p_{n-1}(n)$ of them.

Hence, the total number of partitions of n with each part less than or equal n is $p_{n-1}(n) + 1$ $\square$

We generalize the statement in Example 4 through the following theorem.

Theorem 2.1.

Let $n \geq m > 1$, then $p_m(n) = p(n)_{m-1} + p_m(n - m)$.

Proof. Let $n = a_1 + a_2 + \cdots + a_r$ be a partition of n such that $a_i \leq m$ for all $i \in \{1, \dots, r\}$. Now, observe that either m appears as a part of the partition, or all parts are strictly less than m .

Case 1: The part m appears in the partition. Then the partition has the form

$$n = a_1 + a_2 + \cdots + a_s + m,$$

where each $a_i \leq m$ for all $i \in \{1, \dots, s\}$. Clearly, the number of such partitions is equal to the number of partitions of $n - m$ into parts less than or equal to m , which is $p_m(n - m)$.

Case 2: All parts in the partition are strictly less than m . That is,

$$n = a_1 + a_2 + \cdots + a_r$$

with $a_i \leq m - 1$ for all i .

The number of such partitions is $p_{m-1}(n)$.

Since these two cases are disjoint and together account for all possible partitions of n with parts $\leq m$, we conclude that

$$p_m(n) = p_{m-1}(n) + p_m(n - m),$$

as desired. $\square$

Example 2.5. For $1 \leq j \leq n$, show that the number of partitions of n containing j as a part is $p(n - j)$.

Let λ be a partition of n such that $j \in \lambda$. That is,

$$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k, j)$$

with $\lambda_1 + \lambda_2 + \cdots + \lambda_k + j = n$. Remove one occurrence of the part j from λ . The resulting sum is $n - j$, so the remaining parts form a partition of $n - j$.

Now take any partition μ of $n - j$,

$$\mu = (\mu_1, \mu_2, \dots, \mu_k),$$

and append the part j to get a partition of n :

$$(\mu_1, \mu_2, \dots, \mu_k, j).$$

This partition clearly sums to n and contains j as a part.

3 Ferrers Diagram

Definition 3.1. A graphical representation of a partition of n , where each part is represented by a row of horizontal dots, is called a **Ferrers diagram**.

Example 3.1. Draw the Ferrers diagram for all the partitions of 5 in which each part is less than or equal to 3.

Partition	Ferrers Diagram
3 + 2	
3 + 1 + 1	
2 + 2 + 1	
2 + 1 + 1 + 1	
1 + 1 + 1 + 1 + 1	

Definition 3.2. The **conjugate partition** of a given partition is formed by reading the number of dots in the successive columns of its Ferrers diagram.

Example 3.2. Find the conjugate partition of each partition in example 3.2. and draw their Ferrers diagram.

Conjugate Partition	Ferrers Diagram
$2 + 2 + 1$	<pre> • • • • • </pre>
$3 + 1 + 1$	<pre> • • • • • </pre>
$3 + 1 + 1$	<pre> • • • • • </pre>
$4 + 1$	<pre> • • • • • </pre>
5	<pre> • • • • • </pre>

Notice that these are all the partitions of 5 with at most 3 parts. They are exactly equal in number to the partitions of 5 in which each part is less than or equal to 3.

Theorem 3.1.

If $l_m(n)$ denotes the number of partition of n in which at most m part appear , then

$$p_m(n) = l_m(n).$$

Proof. Every partition of a positive integer n can be represented by a **Ferrers diagram** — an array of left-justified rows of dots, with each row representing a part, and the lengths of rows non-increasing.

Now consider:

- A partition of n into parts $\leq m$ means that in the Ferrers diagram, **no row has more than m dots**.
- A partition of n into at most m parts means that in the Ferrers diagram, **there are at most m rows**.

There is a natural bijection between these two kinds of partitions via the conjugate partition:

- Take a Ferrers diagram of a partition with at most m parts (at most m rows).
- Reflect it along its main diagonal (i.e., transpose the diagram).
- The resulting diagram is a partition in which **each part is $\leq m$** (i.e., at most m columns).

This conjugation is a bijection and preserves the total number of dots (i.e., the integer n). So, the number of such partitions is equal.

Hence,

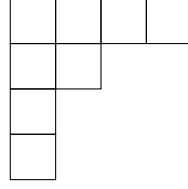
$$p_m(n) = l_m(n). \quad \blacksquare$$

Definition 3.3. A partition is called **self-conjugate** if it is identical with its conjugate.

Example 3.3. Find all the self-conjugate partitions of 8 and, from each one, form a partition into distinct odd parts.

We identify two self-conjugate partitions of 8:

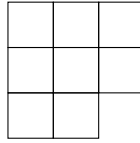
- **Partition 1:** $4 + 2 + 1 + 1$



This corresponds (via hook lengths along the diagonal) to the distinct odd partition:

$$\boxed{7 + 1}$$

- **Partition 2:** $3 + 3 + 2$



This corresponds to the distinct odd partition:

$$\boxed{5 + 3}$$

Theorem 3.2

Let n be a non-negative integer. Then the number of partitions of n into distinct odd parts, *i.e.* $p^o(n)$, is equal to the number of self-conjugate partitions of n .

Proof.

Every self-conjugate partition can be uniquely decomposed into a sequence of L-shaped hooks (a central square on the diagonal and arms going right and downward). Each hook corresponds to a unique odd number (its total length), and since hooks don't overlap and are symmetric, the sizes of the hooks are all distinct and odd. So, given a self-conjugate Ferrers diagram, extract the hook lengths along the diagonal. These hook lengths are distinct odd numbers and sum to n .

Conversely, given a partition of n into distinct odd numbers, we use each odd number as the size of a symmetric hook and arrange them diagonally. This uniquely constructs a self-conjugate Ferrers diagram.

This establishes a one-to-one correspondence between partitions into distinct odd parts and self-conjugate partitions. Thus,

$$p^o(n) = (\text{number of self-conjugate partitions of } n).$$

□

4 Euler's Partition theorem

Before diving into the main theorem of this section let's first consider the following example.

Example 4.1. Find $p^o(6)$ and $p^d(6)$.

- For $p^o(6)$, we list the partitions of 6 into **odd parts**:

$$6 = 5 + 1 = 3 + 3 = 3 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1 + 1$$

Hence, $p^o(6) = 4$.

- For $p^d(6)$, we list the partitions of 6 into **distinct parts**:

$$6 = 6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$$

So, $p^d(6) = 4$ as well.

Thus, we notice that $p^o(6) = p^d(6)$.

Theorem 4.1. (Euler's Partition Theorem)

For any integer n , we have $p^o(n) = p^d(n)$.

*Proof.*¹

We construct a bijection between the set of partitions of n into odd parts and the set of partitions of n into distinct parts.

- Start with a partition of n into odd parts. That is, each part is of the form $2k - 1$, and repetitions are allowed. Group identical parts together. For each part a that appears m times, write m in binary:

$$m = b_0 + b_1 \cdot 2 + b_2 \cdot 2^2 + \cdots + b_r \cdot 2^r, \quad b_i \in \{0, 1\}$$

Replace each group of 2^i identical parts a by a single part of size $2^i \cdot a$. Since the binary digits are either 0 or 1, no resulting part will repeat, ensuring the resulting partition has distinct parts.

- This process is reversible: given a partition into distinct parts, factor each part as $2^i \cdot a$ where a is odd. Then, for each such part, replace it by 2^i copies of a , recovering a partition into odd parts.

Hence, this defines a bijection between the two sets, and we conclude:

$$p^o(n) = p^d(n)$$

□

5 Remarkable Results in Partition theory

This section is dedicated to remarking on some revolutionary ideas that are fundamental in partition theory. However, a rigorous treatment and formal proofs of these results are beyond the scope of this paper.

Euler's Generating Function (1748)

For any integer n , we have

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{j=1}^{\infty} \frac{1}{1-x^j}$$

for $|x| < 1$.

¹The proof presented here is not the one originally given by Euler. Euler's original argument used generating functions to establish the result. In contrast, the proof provided here follows a purely combinatorial approach.

Hardy–Ramanujan Coefficient Formula (1917)

The partition function $p(n)$, as the coefficient of x^n in the generating function, can be written as:

$$p(n) = \frac{1}{n!} \left[\frac{d^n}{dx^n} \left(\prod_{j=1}^{\infty} \frac{1}{1-x^j} \right) \right]_{x=0}$$

This expression gives $p(n)$ exactly via differentiation of the generating function.

Hardy–Ramanujan Asymptotic Formula (1918)

Hardy and Ramanujan found an asymptotic approximation for the partition function $p(n)$. The leading term of their expression is:

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp \left(\pi \sqrt{\frac{2n}{3}} \right)$$

as $n \rightarrow \infty$.

Ramanujan's Congruences (1919)

In 1919, Ramanujan discovered surprising congruence properties of the partition function $p(n)$. For all non-negative integers n , we have:

$$p(5n + 4) \equiv 0 \pmod{5}$$

$$p(7n + 5) \equiv 0 \pmod{7}$$

$$p(11n + 6) \equiv 0 \pmod{11}$$

These congruences revealed deep connections between partitions and modular forms.

6 Conclusion

In conclusion, partition theory shows a fascinating and elegant connection between combinatorial techniques and analytical methods. By utilizing both approaches, partition theory demonstrates how a seemingly simple question can evolve into a complex and profound problem, illustrating the power of mathematical thinking. This field not only bridges two distinct areas of mathematics but also offers insights into how intricate patterns emerge from basic principles, revealing deep relationships within number theory and beyond.

References

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